Monomial f(q) = a*qˆ7 Report

Polynomial

$$f(q) = aq^7$$

Naive

$$g_{\text{naive}}(X) = ax^7$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^7 + 21aq^5\sigma^2 + 105aq^3\sigma^4 + 105aq\sigma^6$$

$$\begin{aligned}\mathrm{Var}(g_{\text{naive}}(X)) = {} & 49a^2q^{12}\sigma^2 + 2352a^2q^{10}\sigma^4 + 40425a^2q^8\sigma^6 + 299880a^2q^6\sigma^8 \\ & + 923895a^2q^4\sigma^{10} + 934920a^2q^2\sigma^{12} + 135135a^2\sigma^{14}\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\text{naive}}) = {} & 49a^2q^{12}\sigma^2 + 2793a^2q^{10}\sigma^4 + 44835a^2q^8\sigma^6 + 315315a^2q^6\sigma^8 \\ & + 945945a^2q^4\sigma^{10} + 945945a^2q^2\sigma^{12} + 135135a^2\sigma^{14}\end{aligned}$$

$$\mathrm{Bias}(g_{\text{naive}}) = 21aq^5\sigma^2 + 105aq^3\sigma^4 + 105aq\sigma^6$$

Unbiased

$$g_{\text{unb}}(X) = -105a\sigma^6x + 105a\sigma^4x^3 - 21a\sigma^2x^5 + ax^7$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^7$$

$$\begin{aligned}\mathrm{Var}(g_{\text{unb}}(X)) = {} & 49a^2q^{12}\sigma^2 + 882a^2q^{10}\sigma^4 + 7350a^2q^8\sigma^6 + 29400a^2q^6\sigma^8 \\ & + 52920a^2q^4\sigma^{10} + 35280a^2q^2\sigma^{12} + 5040a^2\sigma^{14}\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\text{unb}}) = {} & 49a^2q^{12}\sigma^2 + 882a^2q^{10}\sigma^4 + 7350a^2q^8\sigma^6 + 29400a^2q^6\sigma^8 \\ & + 52920a^2q^4\sigma^{10} + 35280a^2q^2\sigma^{12} + 5040a^2\sigma^{14}\end{aligned}$$

Comparison

$$\text{mean gap} = -21aq^5\sigma^2 - 105aq^3\sigma^4 - 105aq\sigma^6$$

$$\begin{aligned}\text{variance gap} = {} & -1470a^2q^{10}\sigma^4 - 33075a^2q^8\sigma^6 - 270480a^2q^6\sigma^8 \\ & - 870975a^2q^4\sigma^{10} - 899640a^2q^2\sigma^{12} - 130095a^2\sigma^{14}\end{aligned}$$

$$\begin{aligned}\text{MSE gap} = {} & -1911a^2q^{10}\sigma^4 - 37485a^2q^8\sigma^6 - 285915a^2q^6\sigma^8 \\ & - 893025a^2q^4\sigma^{10} - 910665a^2q^2\sigma^{12} - 130095a^2\sigma^{14}\end{aligned}$$

$$\begin{aligned}\mathrm{Bias}(g_{\text{naive}})^2 = {} & 441a^2q^{10}\sigma^4 + 4410a^2q^8\sigma^6 + 15435a^2q^6\sigma^8 \\ & + 22050a^2q^4\sigma^{10} + 11025a^2q^2\sigma^{12}\end{aligned}$$